

1. Basic Probability

Sample space and events

The sample space S is the set of all possible outcomes of an experiment.

An event A is any subset of S . Example: rolling a die, $S = \{1,2,3,4,5,6\}$.

Axioms of probability

1. $P(A) \geq 0$ for every event A .
2. $P(S) = 1$.
3. For mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B)$.

Useful rules

Complement: $P(\text{not } A) = 1 - P(A)$

Addition: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Worked example

A card is drawn from a standard deck. $P(\text{heart or king}) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52}$.

2. Conditional Probability

Definition

$P(A | B) = P(A \text{ and } B) / P(B)$, provided $P(B) > 0$.

We restrict the sample space to B and ask what fraction of it also lies in A.

Multiplication rule

$$P(A \text{ and } B) = P(A | B) P(B) = P(B | A) P(A)$$

Independence

A and B are independent if $P(A \text{ and } B) = P(A) P(B)$, i.e. $P(A | B) = P(A)$.

Mutually exclusive events with non-zero probability are never independent.

Worked example

$P(A) = 0.5$, $P(B) = 0.4$, independent: $P(A \text{ and } B) = 0.5 \times 0.4 = 0.2$.

3. Bayes' Theorem

Formula

$$P(A | B) = P(B | A) P(A) / P(B)$$

with $P(B) = P(B | A) P(A) + P(B | \text{not } A) P(\text{not } A)$ (law of total probability).

The base-rate trap

Disease prevalence 1%, test sensitivity 99%, false positive rate 5%.

$$P(\text{disease} | \text{positive}) = 0.0099 / (0.0099 + 0.0495) = 0.167, \text{ only about } 17\%.$$

Rare conditions produce more false positives than true positives.

Tree method

Draw branches for the prior, then for the evidence; multiply along paths, then divide the path you care about by the total of all matching paths.